

Def. Let X_i ($i \in I$) be Topological spaces with Topology $\mathcal{A}_i$, respectively.

Define a Product subbasis on $\prod_{i \in I} X_i$ as:

$$S_p := \left\{ p_i^{-1}[U_i] \mid i \in I, U_i \in \mathcal{A}_i \right\}$$

The Topology $\mathcal{A}_p$ generated by S_p is called a Product Topology on $\prod_{i \in I} X_i$.

Clearly, the product Topology is the smallest Topology s.t. all projections are Continuous.
Moreover, on the Product Space, all projections are Continuous open onto map.

Thm. Let X, Y_i ($i \in I$) be Topological Spaces. Define a map

$$f: X \rightarrow \prod_{i \in I} Y_i : x \mapsto (f_i(x))_{i \in I} \quad \text{where } f_i: X \rightarrow Y_i$$

Then, f is continuous if and only if for all $i \in I$, f_i is Continuous.

Proof If f is continuous, then $f_i = p_i \circ f$ is continuous, being composition of Continuous.

Conversely, let $B \in \beta_p$ be an base element of $\mathcal{A}_p$. Then, by definition,

$$B = \bigcap_{i=1}^n p_i^{-1}[U_i] \quad \text{for some } U_i \in \mathcal{A}_i, \quad i=1, \dots, n.$$

Now,

$$f^{-1}[B] = f^{-1}\left[\bigcap_{i=1}^n p_i^{-1}[U_i]\right] = \bigcap_{i=1}^n (f^{-1} \circ p_i^{-1})[U_i] = \bigcap_{i=1}^n (p_i \circ f)^{-1}[U_i] = \bigcap_{i=1}^n f_i^{-1}[U_i]. \quad \square$$

Projection

Embedding

Lemma.

Let X, Y be Topological Spaces, and let $y_0 \in Y$ be fixed. Define an Embedding

$$E_x : X \rightarrow X \times Y : x \mapsto (x, y_0)$$

Then the E_x is an Embedding.

Proof. Let $U \subseteq X$ and $V \subseteq Y$ be open sets. Then, $U \times V \subseteq X \times Y$ is open set of the Product space. Then, the preimage of E_{y_0} is:

$$E_x^{-1}[U \times V] = \begin{cases} U & \text{if } y_0 \in V \\ \emptyset & \text{if } y_0 \notin V \end{cases}$$

Either case, the preimage is open set in X .

And, given an open set $U \subseteq X$, the image

$$E_x[U] = U \times \{y_0\} = (U \times Y) \cap (X \times \{y_0\})$$

is open set of $X \times \{y_0\}$. Finally, the injectivity is clear, So proved. $\square$

Thm.

Let $f : X \times Y \rightarrow Z$ be a continuous map. Then, for fixed $y_0 \in Y$, the map

$$f_x : X \rightarrow Z : x \mapsto f(x, y_0)$$

is continuous.

Proof. Since $f_x = f \circ E_x$ where E_x is the embedding for fixed $y_0 \in Y$, it is continuous, being the composition of continuous maps. $\square$

Thm. Let X_i ($i \in I$) be Topological spaces, $A_i \subset X_i$. Then,

$$1) \overline{\prod_{i \in I} A_i} = \prod_{i \in I} \overline{A_i}$$

$$2) \left(\prod_{i \in I} A_i \right)^{\circ} \subseteq \prod_{i \in I} A_i^{\circ}, \text{ the equality holds when the product is finite.}$$

$$\text{Proof. } (x_i) \in \overline{\prod_{i \in I} A_i} \Leftrightarrow \forall \mathcal{U} \in \beta_p \text{ with } (x_i) \in \mathcal{U}, \mathcal{U} \cap \prod_{i \in I} A_i \neq \emptyset$$

$$\Leftrightarrow \forall \mathcal{U} \in \beta_p \text{ with } (x_i) \in \mathcal{U}, \prod_{i \in I} \mathcal{U}_i[x_i] \cap \prod_{i \in I} A_i = \prod_{i \in I} \mathcal{U}_i[x_i] \cap A_i \neq \emptyset$$

$$\Leftrightarrow \forall i \in I, \forall \mathcal{U}_i \in \beta_i \text{ with } x_i \in \mathcal{U}_i, \mathcal{U}_i \cap A_i \neq \emptyset$$

$$\Leftrightarrow (x_i) \in \prod_{i \in I} \overline{A_i}$$

(I finite)

$$(x_i) \in \left(\prod_{i \in I} A_i \right)^{\circ} \stackrel{(*)}{\Leftrightarrow} \exists \mathcal{U} \in \beta_p \text{ s.t. } (x_i) \in \mathcal{U} \subseteq \prod_{i \in I} A_i$$

$$\Leftrightarrow \forall i \in I, x_i \in \mathcal{U}_i[x_i] \subseteq A_i$$

$$\Leftrightarrow \forall i \in I, x_i \in A_i^{\circ}.$$

But, if I is Not finite, $(*)$ is broke.